Ji 2020 cSCC — analysis summary

RUN 2 OF 10

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

Patients thoroughly mixed after Harmony (θ=5, chosen on the integration sweep — mixing visually clean, 10x precedent); one connected component (99.7% of cells). Judged on the embedding, not a scalar.

Gate 2 — module geography → GO (non-branching)

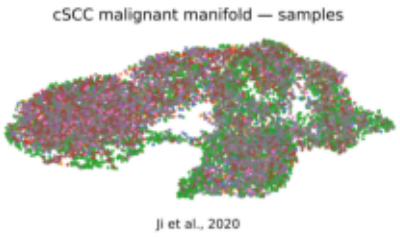
Gast & pSom overlap in the differentiating body (spatial r=+0.10) and both oppose Crest (r=−0.68 / −0.65); along pseudotime Gast declines (ρ=−0.62) and Crest rises to a terminal neural-crest / TSK edge (Crest-high decile pt=0.79). One shared developmental axis, not two competing arms.

Gate 3 — hub cross-repression → NA

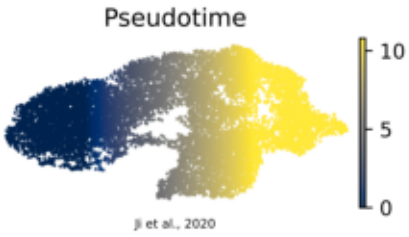
No bifurcation: the only PPT fork is the cycling compartment; the developmental axis is a single fold-free curve. Organized gene-level Gast↔pSom cross-repression (r=−0.53) IS present on the hub, but with Gast/pSom overlapping and no arm resolution this is the non-branching-with-cross-repression case (task §1), not a bifurcation.

STEP FIGURES

Manifold — samples



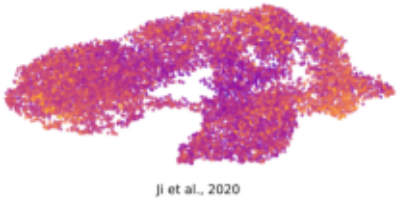
Pseudotime



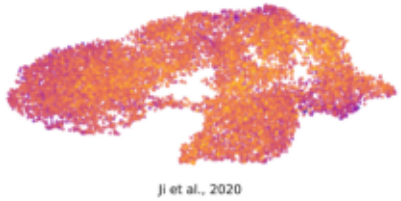
Gastrulation



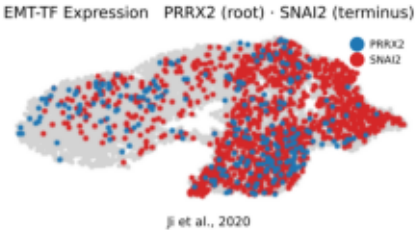
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous squamous cell carcinoma (Ji et al., 2020), 10x — 10,504 malignant cells from 8 patients (P3/P5 dropped below the count cliff).

Verdict.
The embryonic pattern reproduces as a NON-BRANCHING axis: a Gast/pSom-high differentiating origin grades to a neural-crest-high basal/TSK terminus (Crest the inverse/terminal readout). No developmental bifurcation — the sole fork is proliferation. Linear geometry; Gate 3 not applicable. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs PRRX2 (root-end, near floor) → SNAI2 (terminus).

Parameter choices

Malignant load	10,598 cells / 10 patients (publication calls)	**Sample threshold**	≥103 (geom mean of 268→40 cliff); 8/10 kept → 10,531 cells
Island removal	UMAP radius-graph (6× median NN); ~27 detached → 10,504	**Integration**	Harmony θ=5; n_pcs=20; UMAP spread=1.3
Trajectory	scFates curve (ElPiGraph) N=10 μ=100; 2 tips, 0 forks; fold-free (SA 3.0)	**Root**	Gast/pSom-high, Crest-low curve tip
Hub (context)	Mixed+Diff core; gene-level Gast↔pSom r=−0.53	**EMT-TFs**	SNAI2 (terminus); PRRX2 root-end near floor